

Based on application, it is segmented into consumer electronics, retail, government, enterprise, industrial, commercial (hospitality, transport and gaming) and others (medical, military and training).

Overview of various multi-touch technologies

Touch panels are becoming more common in our daily lives, in cell-phones, ATMs, kiosks, ticket vending machines, etc to enable user interaction. Each device uses a different technology to meet the demand.

Electrostatic systems. Projected

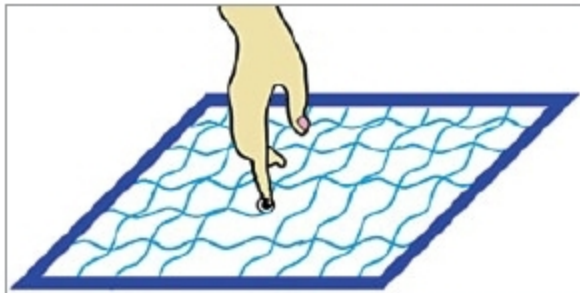


Fig. 9: Acoustic surface

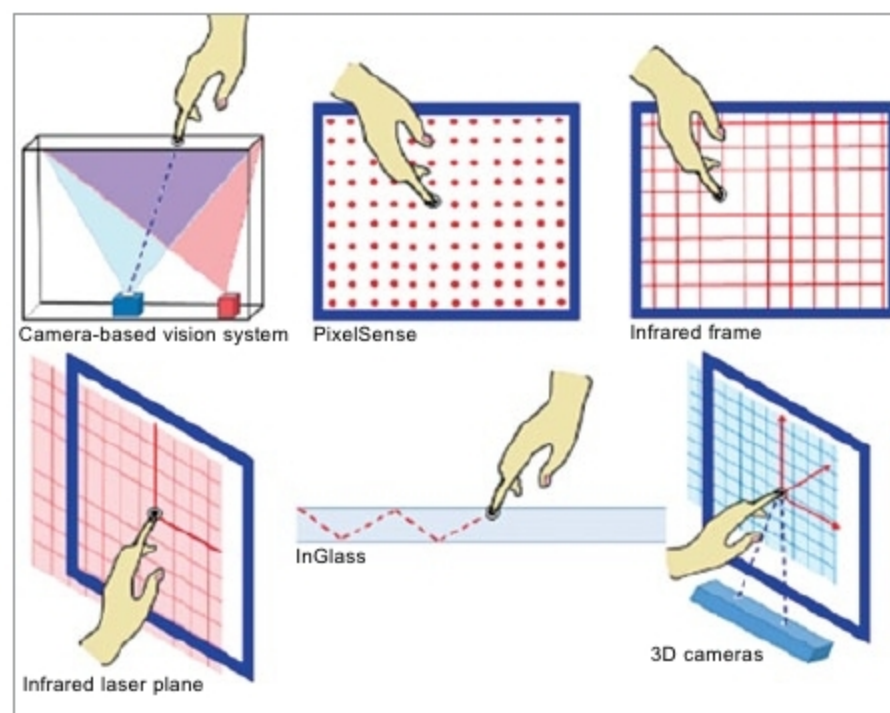


Fig. 10: Various optical systems

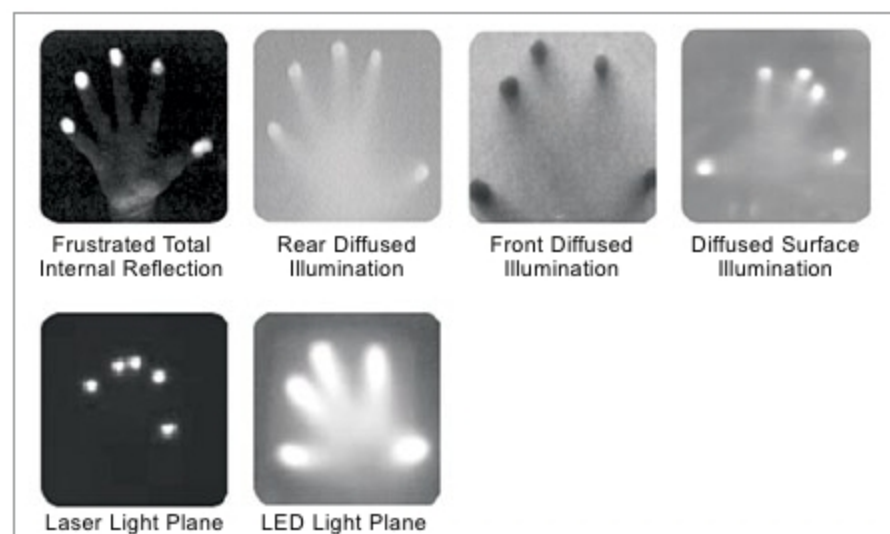


Fig. 11: Clarity and precision of detection of objects for all techniques of touch technology (Credit: sethsandler.com/multitouch)

capacitance technology is a sensitive sensor that can sense through a 6mm-thick glass (insulator), coated with a conductor like indium-tin-oxide. When the user touches the surface of the screen, electricity is conducted through the finger causing a measurable change in capacitance. This change in capacitance creates a distortion in the screen's electrostatic field.

It is mainly used in control panels, digital signage, industrial automation, gaming machines and so on. Touchscreens manufactured with this technology give excellent resistance to surface contaminants and liquids, a long life span with more simultaneous touch points, use less power, and are more reliable and durable. However, this technology requires initial tuning of devices, so it is expensive, and we can use only conductive elements (gloves cannot be used).

Force touch systems. 3D touch technology is the same as projected capacitance, and sensors can measure small changes in distance between the cover glass and back-light due to force of the finger. 3D touch technology is mainly used in all iOS devices. These are extremely strong, highly reactive and immune to ambient light. But this technology is limited to object recognition—similar to projected capacitance, where gloves cannot be used. These touchscreens are not available in large formats.

Pressure-based systems. Resistive touch technology is mainly used in aviation, military, broadcast and education devices. Two conductive layers are separated by spacers. When the

user touches one conductive layer, pressure creates contact between the two layers, causing current to flow. So, usage of gloves by the user does not affect the flow of current.

This technology is easy to manufacture and costs low if size of the screen is normal, but increases exponentially for large sizes. Touchscreens made using this technology are more stable, resistant to screen contaminants and immune to ambient light. But these have poor object recognition, poor clarity and poor multi-touch capability.

Acoustic systems. Acoustic surfaces are mainly used in point-of-information kiosks, hospitality, ticket sales (transportation), multimedia payphones, gaming, lotteries and amusement. Sides of the monitor have a series of piezoelectric transducers and receivers that create an invisible grid of ultrasonic waves on the surface. When a user touches the panel, the finger absorbs part of the wave that is produced, which allows the receiving transducer to locate the touchpoint(s).

Acoustic surface touch panels address the limitation of poor clarity in projected capacitance and resistive touch. These also ensure better durability, high stability, longer life span and scratch-resistance. The remarkable feature of this technology is that the screen is tested at over fifty million touches.

It is not recommended to use acoustic products in open environments. Also, the screen cannot be completely sealed, is immune to low weather and surface contaminants cause dead spots on the screen.

Optical systems. Optical systems are cost-effective and the easiest to create. Fig. 10 shows the various optical systems, wherein each system uses different optical techniques such as frustrated total internal reflection (FTIR), rear diffused illumination (RDI), front diffused illumination (FDI), diffused surface illumination (DI), laser light plane (LLP) and LED light plane (LED-LP). Clarity and precision of detection of objects for each technique can be seen in Fig. 11.

Camera-based vision systems. These have two or more IR cameras placed on top of a display, facing the